

■ PYTHON CLASS 3

LOOP QUEST

5 beginner-friendly missions to make Python repeat, filter and process

THE BIG IDEA

You already taught Python how to store information, take input, calculate and make decisions.

Now give Python a new superpower:

REPEAT THE WORK FOR YOU.

Your mission path

REPEAT -> CONTROL -> FILTER -> PROCESS DATA -> BUILD

Friendly rule: These are not trick questions. Start simple, get it working, then try the ■ challenge.

Before you start

- Write the code yourself.
- Run every program with a few different values.
- Predict the output before you press Run.
- If something breaks, read the error and try again.

Submission: Save all 5 programs in one folder or GitHub repository. Add a screenshot of a successful run for each mission.

Story

Imagine a tiny robot that can take one step at a time. You want it to walk forward a certain number of steps.

Your mission

Ask the user for the number of steps and print a message for every step.

```
Enter steps: 5
```

```
Step 1
```

```
Step 2
```

```
Step 3
```

```
Step 4
```

```
Step 5
```

Think first

How many times should the loop run? What number should change each time?

Hint: This is a perfect job for `for + range()`.

■ Level Up

Change the message so it says **"Robot moved to step 1"**, **"Robot moved to step 2"**, etc.

■ Done when...

Your program works for 3, 5 and 10 steps without changing the loop itself.

Scan a range and keep only the numbers you want.

Story

A little scanner checks numbers from 1 to 30. It is only interested in **even numbers greater than 10**.

Your mission

Print every number from 1 to 30 that is both even and greater than 10.

Expected output:

```
12
14
16
18
20
22
24
26
28
30
```

Think first

You need two ideas working together: repeat through the numbers, then test each number.

Hint: Build the loop first. Then use `% 2 == 0` for even numbers. Finally combine the rules with `and`.

■ Level Up

Ask the user for the maximum number instead of always stopping at 30.

■ Done when...

You can change the maximum number and your program still filters correctly.

Story

You're designing a game power-up meter. Every level gives you one more multiple of a chosen number.

Your mission

Ask the user for a number and how many multiples to generate.

```
Enter base number: 4
```

```
Enter how many: 5
```

```
4
```

```
8
```

```
12
```

```
16
```

```
20
```

Think first

The pattern is not random. What happens to the result each time the loop repeats?

Hint: If the base is 4, try 4×1 , 4×2 , 4×3 ...

■ Level Up

Let the user choose the base and the number of multiples. Test with 7 and 6.

■ Done when...

You can change both inputs and generate the new pattern without rewriting the program.

Story

You have a list of student scores. Your job is to quickly find the students who reached the target score.

Starting data

```
scores = [42, 78, 91, 33, 67, 85, 49, 73, 58, 88]
```

Your mission

Use a loop to print only the scores that are **70 or above**.

Expected output:

```
78
91
85
73
88
```

Then upgrade it

Also print how many scores were 70 or above.

Hint: A loop can check each score. A variable such as `count = 0` can keep track of how many match the rule.

■ Why this matters later

This is your first small taste of **filtering data**: look at many values, keep the ones that match a rule.

Story

A weather app collected 10 daily temperatures. Build a tiny analyzer that scans the data and reports useful results.

Starting data

```
temperatures = [31, 34, 29, 36, 33, 27, 38, 35, 30, 32]
```

Your mission

Using a loop, produce these results:

- 1 Print every temperature.
- 2 Print temperatures that are **35 or above**.
- 3 Count how many days were 35 or above.
- 4 Count how many days were below 30.

Expected output structure

```
All temperatures:
31 34 29 36 33 27 38 35 30 32

35 or above:
36
38
35

Days at 35+:
3

Days below 30:
2
```

■ Final Level Up

Calculate the total and average temperature using a running total. Build the logic yourself.

Think before coding: What information must your program remember while the loop is running?

■ QUEST COMPLETE

Show what you built.

Before you submit:

- All 5 programs run
- You tested different values
- You used a for loop where required
- You predicted outputs before running
- You can explain your own code
- You attempted at least one ■ challenge

What you should now be able to see

A repetitive task
↓
A loop
↓
A condition / calculation
↓
Many values processed
↓
Useful result

One final question:

Why is a loop more powerful than writing the same instruction again and again?

Answer in your own words. No textbook definition needed.

The bigger journey

Loops are one small building block. Soon we will use them to work with richer collections of data, reusable functions, files, NumPy and eventually Machine Learning.

Learn → Practice → Build → Explain